Multimodal Industrial Anomaly Detection via Attention-Enhanced Memory-Guided Network

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***Abstract*—Anomaly detection is a key technology in quality control for automated production lines. Currently, 2D-based anomaly detection methods fail to identify geometric structure anomalies in products. To address this limitation, this paper proposes a multimodal anomaly detection model using 3D point clouds and RGB images. To ensure the single-domain inference capability of each modality, we design an attention-enhanced dual memory bank to separately store local point cloud features and RGB features. The attention mechanism enhances the informativeness and discriminability of the feature descriptors, significantly improving the data quality in the memory bank. During the inference phase, the local point cloud features in the dual memory bank guide the RGB features in calculating anomaly scores in the 2D modality. This memory-guided approach strengthens the correlation between information across different modalities. Moreover, to improve the overall segmentation precision of the model, we propose an anomaly scoring scheme based on a weight map of signed distance values. The final anomaly detection results are obtained by integrating the advantages of point cloud data in geometric structure anomaly detection and RGB data in color anomaly detection. Extensive experiments demonstrate that the proposed method achieves superior segmentation precision compared to other advanced methods on the MVTec 3D-AD and Eyecandies datasets.**

***Index Terms*—Anomaly detection, point cloud, signed distance values, multimodal learning.**

I. Introduction

With the development of artificial intelligence technology, anomaly detection (AD) in multimedia data has been widely applied in fields such as medical image analysis [1], [2], industrial defect detection [3], [4], security monitoring [5], [6], and autonomous driving [7]. AD is a critical aspect of industrial quality control, aiming at identifying anomalies in a product to ensure the quality and reliability of the product [8]. Since abnormal samples are scarce and difficult to collect in real-world manufacturing, unsupervised methods are primarily employed to identify anomalies within industrial environments. Such methods are usually trained only on a large set of normal samples and the target samples are detected in the inference stage. Currently, the majority of industrial anomaly detection methods [9], [10], [11], [12], [13] are still based on RGB image analysis. However, in many industrial scenarios, relying only on color images makes it difficult to accurately identify anomalies [14]. Different lighting conditions may lead to false detections, while anomalies of geometric structure types may not present variations in color. Therefore, effectively leveraging the synergy between RGB and 3D information is crucial for improving the AD performance. However, using multimodal data for AD also presents several challenges. First, different modalities may provide conflicting information. An RGB image may show a color anomaly, while the 3D data indicates that the geometric structure is normal. Such inconsistencies can make it difficult for the model to accurately identify anomalies. Moreover, different modalities are affected by different factors. RGB images can be influenced by lighting conditions, while 3D data may suffer from noise or missing points. These quality differences may cause multimodal data to fail in accurately representing anomalies.

The recently released MVTec 3D-AD dataset [15], as a benchmark resource, has greatly advanced the field of multimodal anomaly detection [16], [17], [18]. AST [19] reduces the overgeneralization of anomalous samples by using an asymmetric teacher-student network, but it relies on the role of RGB information and does not fully leverage the detection advantages of 3D data, which limits its detection performance. To leverage the inferential advantages of each modality, MIAD [20] achieves anomaly detection by comparing the differences between the features mapped through two cross-modal mapping functions and those learned by the feature extractor. In addition, MIAD obtains three variants, MIAD-M, MIAD-S, and MIAD-T, by pruning the network layers of the feature extractor. However, they require designing complex training goals separately for each modality. Memory bank-based methods have gained significant attention, as these approaches only need to store features of normal samples during training without requiring complex training designs. BTF [14] attempts to improve the utilization of multimodal data by connecting features from two modalities into a memory bank. Nevertheless, directly connecting multimodal features overlooks potential interference between them, which can significantly impact detection performance. M3DM [21] adopts a hybrid fusion scheme to alleviate the problem of interference among modalities. It constructs three memory banks for each modality and fusion feature separately to

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strengthen the single-domain inference capability of the model. Nonetheless, this approach results in substantial memory costs. Shape-guided [22] reduces memory usage by constructing a dual memory bank with two expert models. However, the direct guidance approach between the expert models does not ensure the quality of the features stored in the memory bank, which directly impacts the final detection performance.

To alleviate the issue of data quality in the memory bank, we use attention-enhanced memory guidance to construct a dual memory bank. Specifically, we use attention mechanism to strengthen connections between multimodal data. The 3D features guide the attention-enhanced RGB features, together forming the dual memory bank. Although 3D features provide geometric structural information about objects, in RGB images, color variations in certain regions may exhibit more anomaly potential than changes in 3D geometry. The attention mechanism effectively enhances the feature representation of these regions. The design provides two advantages. First, the attention mechanism enhances the informativeness and distinctiveness of feature descriptors, significantly improving data quality within the memory bank. Second, it increases pixel-level alignment accuracy between modalities, which is crucial for detecting subtle anomalies. Meanwhile, we design a weighted map anomaly scoring scheme based on signed distance values for point cloud data and a cosine similarity scoring scheme for RGB data. The above design ensures the single-domain inference capability of each modality. For the final anomaly localization, we first map the RGB score distribution to align with the 3D score distribution, and then select the larger value at each pixel location as the final pixel-level anomaly score. The above operation effectively combines the detection advantages from different modalities. It is noteworthy that our method shows superior anomaly detection and segmentation performance on the MVTec 3D-AD benchmark, maintaining competitive results even at low false positive rate (FPR) settings.

The contributions of this study can be summarized as follows:

- This study proposes an unsupervised multimodal industrial anomaly detection method based on an attention-enhanced memory-guided network. The method effectively combines the advantages of RGB and 3D data for anomaly detection, addressing the performance limitations inherent in single-modality approaches.
- This study employs attention mechanism to improve the data quality in the memory bank, thereby enhancing the pixel-level correspondence accuracy between multimodal data during the memory-guided process.
- An anomaly scoring scheme based on a weight map of signed distance values is proposed in this paper, which effectively solves the problem of poorly localized regions with weakly expressed anomalies. This greatly improves the final detection and segmentation precision of the model.
- To the best of our knowledge, the proposed method achieves state-of-the-art performance in anomaly segmentation on the MVTec 3D-AD dataset and Eyecandies dataset. Even under high precision requirements, it still demonstrates leading segmentation results.

The rest of this paper is organized as follows: Section II provides a review of related works. Section III presents our method. Section IV conducts the experiments. Finally, Section V concludes the paper and discusses the limitations of the proposed methods.

II. Related Work

A. 2D Industrial Anomaly Detection

The public availability of the MVTec AD dataset [23] has significantly contributed to the development of 2D industrial anomaly detection research. Common approaches include reconstruction-based [24], [25], [26], [27] and feature embedding-based [28], [29], [30], [31] methods. For reconstruction methods, regions where the reconstructed image differs significantly from the input image are considered as the location of the real anomalies. On this basis, in order to enable the model to learn anomalous features during training, researchers employ data augmentation techniques [32], [33] that introduce pseudo-anomalies into normal images. Knowledge distillation [34], [35] is based on a teacher model to train student networks for image reconstruction or feature extraction. Anomaly localization is achieved by comparing the representation differences across the networks. Feature embedding methods aim to transform the original data into a new representation space that facilitates anomaly detection. Normalization flows [28] employ invertible transformations to map data features to a normal distribution. This process allows normal and anomalous samples to exhibit distinguishable probability density distributions in the embedding space. Visual-language models identify anomalies by calculating the similarity between images and textual descriptions [36]. The use of learnable prompts can effectively enhance model performance [37], [38]. Methods based on pre-trained networks and memory banks [39] often face memory issues and computational burdens. PatchCore [12] stores local features of normal samples in a memory bank and then locates anomalies by comparing the similarity between the target sample and the stored features. Inspired by this approach, we applied the memory bank method to multimodal anomaly detection tasks and achieved excellent performance.

B. Multimodal Industrial Anomaly Detection

Research [14] shows that effectively combining RGB and 3D information leads to significantly better anomaly detection performance than using only RGB data. With the release of the MVTec 3D-AD dataset [15], 3D industrial anomaly detection has become a new research focus. The study that published the dataset also offered baseline models based on GAN, AE, and variational models, each trained on 3D data represented by voxel grids or depth maps. 3D-ST [18] proposes a student-teacher framework, where the teacher network encodes local geometric descriptions from patches to locate geometric anomalies in point cloud data. BTF [14] examines the role of 3D information in anomaly detection tasks, revealing the potential advantages of combining RGB and 3D information for anomaly localization.

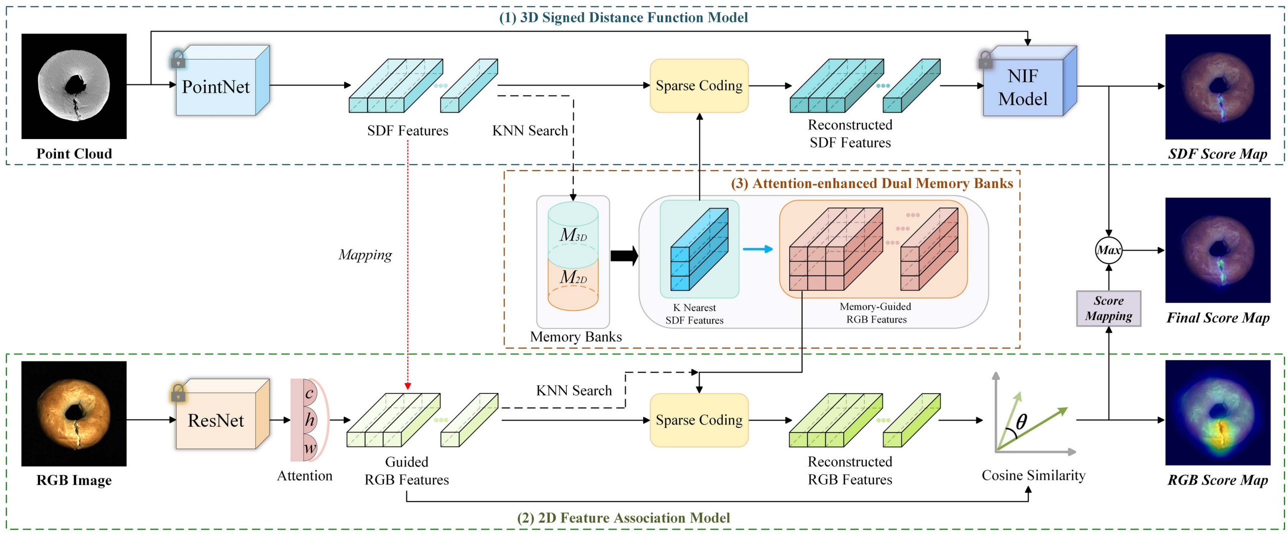


Fig. 1. Diagram of the inference process. The proposed method consists of three main parts: (1) **3D signed distance function model** predicts the signed distance values of the local features in the point cloud, generating an anomaly score map for the 3D data. PointNet is a pre-trained 3D feature extractor, and the NIF model is a multi-layer perceptron used to represent local 3D shapes. (2) **2D feature association model** uses Wide-ResNet50-2 to extract RGB features. Multi-dimensional attention enhances the detailed representation of the RGB feature map. The guided RGB features are obtained through the mapping of the SDF features. Finally, an anomaly score map for the RGB data is produced. (3) **Attention-enhanced dual memory banks** store the normal features of each modality during the training phase. M_{3D} stores local 3D features, while M_{2D} retains the attention-enhanced and guided RGB features. The quality of the stored data impacts the final model's performance.

However, if the advantages of multimodal data are not fully integrated, performance may be lower than when using only RGB data. For example, AST [19] and EasyNet [40] primarily rely on RGB features to identify anomalies, while the contribution of 3D features is limited. In these cases, directly combining RGB and 3D features can reduce model performance, as 3D features may be viewed as interference to RGB features. Therefore, effectively integrating the advantages of multimodal data is crucial for improving anomaly detection performance. DRAIN [41] employs a lightweight MLP and a dual-attention information entropy fusion strategy within a dual-reconstruction network to enhance feature extraction and integration. Existing methods fail to fully ensure the single-domain inference capability of each modality. Inspired by this, we design an attention-enhanced memory-guided network that retains the advantages of single-modal detection while improving the quality of data in the memory banks. The multi-dimensional attention mechanism [42] with its representational advantages in modeling across three dimensions, enables the model to better distinguish between normal and anomalous instances. In addition, the correlation between the information in different modalities is strengthened by utilizing 3D local features to guide the RGB features. Finally, efficient integration of the advantages of each modality is achieved by mapping the anomaly scores to the same distribution and then by taking the maximum value for each pixel.

III. Method

The proposed method consists of three main components: a 3D signed distance function model, a 2D feature association model, and an attention-enhanced dual memory bank. Notably, the attention-enhanced dual memory bank is constructed during the training phase and utilized in the inference phase to calculate anomaly scores across different modalities. The construction processes of the memory banks for each modality are described separately in Sections III-A and III-B, while their roles during inference are detailed in Section III-C. The anomaly scoring scheme based on a weighted map further helps the model achieve more accurate anomaly segmentation results, as described in Section III-C2. The inference diagram of the proposed method is shown in Fig. 1. The key notations and variables used in this paper, along with their definitions, are detailed in Table I.

TABLE I
Reference Table of Key Notations and Variables in This Paper

Notation	Define
f	Local geometric feature vector
ψ	Neural implicit function model
q	Query point
Q	Query point set
s	Signed distance value
t_i	Nearest neighbor position
$t_i^{'}$	Pulled query point position
Δs	Gradient of the signed distance value
M_{3D}	3D feature memory bank
M_{2D}	2D feature memory bank
$\tilde{f}$	3D feature vector of the test sample
$\ddot{f}$	2D feature vector associated with $\tilde{f}$
$\hat{f}$	Reconstructed representation of $\tilde{f}$
$\bar{f}$	Reconstructed representation of $\ddot{f}$

A. 3D Signed Distance Function Model

The 3D signed distance function model detects anomalies by focusing on local geometric features. As shown in Fig. 2,

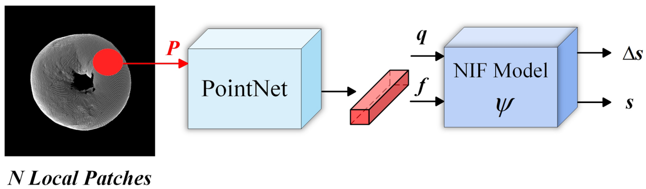


Fig. 2. Diagram of the 3D signed distance function model.

PointNet [43] and neural implicit functions (NIF) [44] are used to capture 3D local structural information. The design of the model is driven by two core principles: anomalies are typically reflected in local rather than global features, and learning local geometric features offers better scalability.

Concretely, the entire point cloud is divided into multiple 3D local patches, and each patch undergoes local feature learning. Each patch is composed of 500 points sampled from the point cloud, and PointNet is used to extract their feature vectors f, which represent the local geometric structure. The NIF model of the multilayer perceptron architecture is denoted as ψ, which can be used for the anomaly detection task after the below training. Following the approach by Ma et al. [45], a set of query points $Q = \{q_1, q_2 \cdots q_i\}$ is sampled near the surface of each 3D local region. These query points, along with the feature vectors f extracted by PointNet, are input into the NIF model ψ. The model then predicts their signed distance values $\{s_1, s_2 \cdots s_i\}$. The process of predicting the signed distance s of each query point q relative to the local surface is represented as:

$$q_i \in Q \xrightarrow{\Psi, f} s = \psi\left(q; f\right) \tag{1}$$

Each pair $\{\psi, f\}$ forms a signed distance function (SDF) that measures the local surface geometry of the point cloud. The NIF model predicts the signed distance s for each query point q through the process described above. ψ minimizes the distance between the pulled points and the actual surface points during the training process achieved by the pulling loss, which enables the model to learn accurate local geometric features. The loss is defined as follows:

$$L_{pull} = \frac{1}{I}\sum_{i=1}^{I} \|t'_i - t_i\|_2^2 \tag{2}$$

where I represents the total number of query points. t_i is the actual nearest neighbor point of q_i in the point cloud P. t'_i is the pulled query point's position. The pulling process is expressed by the equation below:

$$t'_i = q_i - s \times \frac{\Delta s}{\|\Delta s\|^2} \tag{3}$$

where Δs represents the gradient, which is the derivative of s with respect to the position of the query point q.

In summary, through the above process, the model gains the ability to characterize normal local geometric features, and it can effectively identify abnormal features in the inference phase. After training the 3D SDF model, all local region feature vectors $\{f\}$ are stored in a memory bank M_{3D}. This process implicitly encodes all "normal" local geometric representations.

B. 2D Feature Association Model

The 2D feature association model relies on the interrelationship between 3D data and corresponding RGB images. Based on the previously developed 3D SDF model, the mapping relationship between SDF and RGB features is established using pixel-level coordinate information from the data. Specifically, the goal of the 2D feature association model is to construct an SDF-guided memory bank, denoted as M_{2D}, for reconstructing the normal patterns of RGB features. As shown in Fig. 3, for each 3D patch feature outputted by PointNet (in blue), we map the coordinates of 500 points within its local region onto a 2D plane to locate the associated RGB region (marked by a dashed red circle). Additionally, we expand the associated RGB region by a 2-pixel neighborhood (yellow area surrounding the red circle). This strategy effectively enhances the model's ability to capture color variations and texture details. Ultimately, multiple RGB feature vectors (in light pink) associated with each SDF feature are stored in the memory bank M_{2D}. M_{3D} and M_{2D} together form the attention-enhanced dual memory bank.

Fig. 4 illustrates how the input feature maps are processed through the three separate paths of the attention module. In the top path, the feature map is first rotated 90° counterclockwise around the H-axis to form an intermediate feature map $\breve{F}_W$. To simulate deep dependencies between channels and spatial dimensions, $\breve{F}_W$ undergoes a squeeze transformation to obtain the aggregated feature map $\hat{F}_W$. Subsequently, $\hat{F}_W$ is processed through an excitation transformation to capture spatial feature interactions, resulting in the feature weight map $\tilde{F}_W$. These feature weight maps are then processed by a sigmoid function to yield the attention weight map A_W, which is combined with $\breve{F}_W$ to generate the enhanced feature map F'_W. Finally, F'_W is rotated 90° clockwise around the H-axis to match the shape of the original input feature map, resulting in the final feature map F''_W. The process described above is represented by the following equation:

$$\breve{F}_W = PM_H(F) \tag{4}$$

$$\hat{F}_W = T_{sq}(\breve{F}_W), \tilde{F}_W = T_{ex}(\hat{F}_W) \tag{5}$$

$$A_W = \sigma(\tilde{F}_W), F'_W = A_W \otimes \breve{F}_W \tag{6}$$

$$F''_W = PM_H^{-1}(F'_W) \tag{7}$$

where $PM_H(\cdot)$ represents a 90° counterclockwise rotation around the H-axis, and $PM_H^{-1}(\cdot)$ represents the corresponding inverse rotation. $T_{sq}(\cdot)$ and $T_{ex}(\cdot)$ denote the squeeze and excitation transformations, respectively, and σ represents the sigmoid activation function. The middle path operates similarly to the top path. The bottom path focuses on channel interactions. Initially, the feature map F is mapped to $\breve{F}_C$. Then, $\breve{F}_C$ undergoes squeeze and excitation transformations to form the aggregated feature map $\hat{F}_C$ and the feature weight map $\tilde{F}_C$. Subsequently, $\tilde{F}_C$ is processed by the sigmoid function to obtain the channel-specific attention weight map A_C, which scales $\breve{F}_C$ to generate the enhanced feature map F'_C. Finally, F'_C is remapped to form F''_C. The above process can be described by the equations below:

$$\breve{F}_C = IM(F) \tag{8}$$

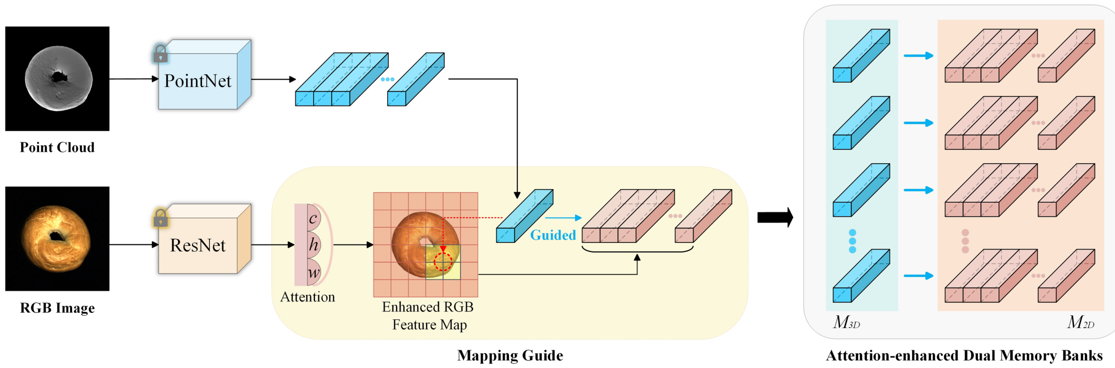


Fig. 3. Attention-enhanced dual memory banks. The 3D SDF model stores feature vectors $\{f\}$ of all 3D patches in the memory bank M_{3D}. Each patch corresponds to a specific region in the RGB feature map, generating a dedicated dictionary. All RGB dictionaries guided by patch information are stored in the memory bank M_{2D}. M_{3D} and M_{2D} together form the dual memory banks utilized during inference. Notably, the RGB feature map is refined through multi-dimensional attention, which significantly improves the data quality in the memory banks.

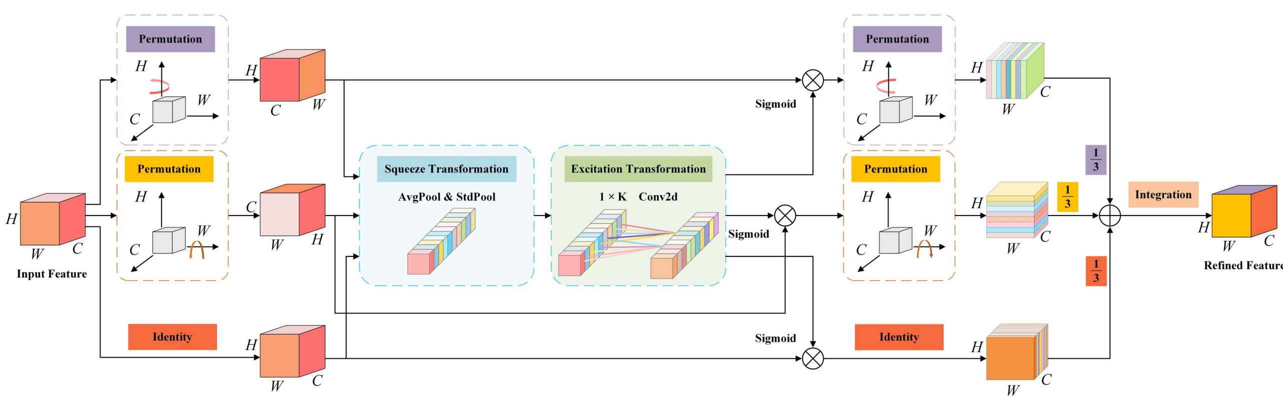


Fig. 4. Illustration of the multi-dimensional attention mechanism. The architecture contains three branches. The top branch and the middle branch are concerned with capturing feature interactions in the spatial dimensions W and H, respectively. While the bottom branch focuses on capturing interactions between channels. $\otimes$ denotes the element-wise multiplication operation and $\oplus$ denotes the element-wise summation operation. This design strengthens the refined representation of features by efficiently integrating information from different dimensions.

$$\hat{F}_C = T_{sq}(\breve{F}_C), \tilde{F}_C = T_{ex}(\hat{F}_C) \tag{9}$$

$$A_C = \sigma(\tilde{F}_C), F'_C = A_C \otimes \breve{F}_C \tag{10}$$

$$F''_C = IM(F'_C) \tag{11}$$

where $IM(\cdot)$ refers to the identity mapping function. The multi-dimensional attention mechanism enhances the learning and representational capacity of RGB features through direct averaging aggregation across three branches, thereby significantly improving the pixel-level correspondence precision between 3D and RGB features. Furthermore, the enhancement of data quality within the memory bank positively contributes to the model's performance.

C. Inference Phase of the Proposed Model

Based on the above process, we utilize the 3D signed distance function model, the 2D feature association model, and the attention-enhanced dual memory bank to identify anomalies in the test sample x. The schematic of the inference phase is presented in Fig. 1, with the following steps:

1) Feature Extraction: PointNet and Wide-ResNet50-2 are used to extract all patch-level SDF and RGB features from the test sample x. The SDF feature vectors are denoted as $\{\tilde{f}\}$. The multidimensional attention mechanism models attention across channels, height, and width, enhancing the expressive power of the RGB feature map. Pixels related to at least one SDF feature in $\{\tilde{f}\}$ are considered the foreground of the RGB image. The RGB feature vectors associated with the SDF features after attention are denoted as $\{\dddot{f}\}$.

2) Inference of the 3D Signed Distance Function Model: For each SDF in $\{\tilde{f}\}$, find 10 nearest neighbors in the M_{3D} to form a feature dictionary and perform sparse coding to obtain approximate features $\{\hat{f}\}$. The purpose of sparse coding is to represent the input features $\{\tilde{f}\}$ as a sparse linear combination of the nearest neighbor features. This allows the original features $\{\tilde{f}\}$ to be approximated by a few neighboring features, achieving a sparse representation. Since sparse coding uses normal features in M_{3D} as basis vectors, the sparse representation can accurately describe normal features, facilitating the distinction between normal and anomalous features. For each patch of x, compute the signed distance values of all 3D points $\{\hat{q}\}$ using the patch-reconstructed $\{\hat{f}\}$, denoted as $s = \psi(\hat{q}; \hat{f})$. Since in the training phase, the NIF model has learned to represent normal 3D local features. In the testing phase, a signed distance value of 0 predicted by the NIF model indicates normal features, while non-zero signed distance values are considered to be an indication of anomalies. Both positive and negative values of s represent anomalies, with the absolute value of s indicating the extent of the anomaly. Therefore, the absolute value of s is considered as the score to quantify the anomaly. Our aim is to detect all anomalies. If the difference in the absolute values of s is too large, smaller s values may not be accurately localized. This can affect the model's final detection and segmentation performance. To overcome this problem, we design an anomaly scoring scheme based on a logarithmic weighting map of s. The scheme is described in more detail below:

First, the absolute value of the signed distance s is taken for all 3D points $\{\hat{q}\}$. Next, the logarithmic function $log(1{+}x)$ is used as a smoothing function to compute the weights. The logarithmic function exhibits rapid growth for small x and slower growth for large x, which reduces the magnitude differences of s and enhances the focus on smaller s values. The above process can be described as:

$$s_L = |s_L| \tag{12}$$

$$weight = log\left(1 + s_L\right) \tag{13}$$

where L represents the index variable for traversing each s value. Then traverse the s value at each position to update the score map and weight map. When updating the score map, if the s value at a position is smaller than the value at the corresponding position in the score map, the score map is updated to the current s value. This ensures that the s value at each position represents the distance to the nearest surface. Meanwhile, to consider the effects of all SDFs in $\{\tilde{f}\}$, the weight map is updated by accumulating logarithmic weights about s values for all the same positions. Additionally, normalize the weight map to balance the weights. The specific computational procedure described above is shown below:

$$s\_map_L = \min\left(s\_map_L, s_L\right) \tag{14}$$

$$w\_map(p) = \sum_{n=1}^{N} weight_n(p), p \in P \tag{15}$$

$$w\_map = \frac{w\_map}{\sum w\_map + \varepsilon} \tag{16}$$

where $w\_map(p)$ denotes the value of the weight map at pixel location p. P indicates all pixel positions. N means the number of patches containing pixel position p. ε is a small constant to prevent division by zero. The final score map is obtained by multiplying the original score map with the weight map. The description above is shown below:

$$s\_map_{sdf} = s\_map \otimes w\_map \tag{17}$$

where $\otimes$ represents element-wise multiplication. s_map_{sdf} represents the anomaly score map obtained from the SDF model using 3D data as input.

3) Inference of the 2D Feature Association Model: For all the associated SDFs used to compute the approximate feature $\{\hat{f}\}$ in M_{3D}, the concatenation of their associated RGB dictionaries is taken in M_{2D} to form a memory-guided RGB dictionary, denoted as $\hat{D}$. For the RGB feature vectors in $\{\dddot{f}\}$, identify their 5 nearest neighbors from $\hat{D}$ and obtain their sparse representations, denoted as $\{\bar{f}\}$. Since cosine similarity focuses on the direction rather than the magnitude of vectors, it better handles the similarity of sparse vectors. Here, we use cosine similarity to calculate the similarity between $\{\dddot{f}\}$ and $\{\bar{f}\}$, with lower similarity indicating a higher degree of anomaly. The calculation process for the anomaly score map is as follows:

$$s\_map_{rgb} = 1 - cos(\dddot{f}, \bar{f}) \tag{18}$$

$$cos\left(\dddot{f}, \bar{f}\right) = \frac{\dddot{f} \cdot \bar{f}}{\left\|\dddot{f}\right\| \left\|\bar{f}\right\|} \tag{19}$$

where s_map_{rgb} represents the anomaly score map obtained from the 2D feature association model using 2D data as input.

4) Score Map Alignment: To fully utilize the inference advantage of each modality, we map the RGB score distribution to the SDF score distribution. Then the anomaly scores in both modalities are compared pixel by pixel. The larger value is taken as the final anomaly response. To obtain the mapping relationship between the score distributions of different modalities, we simulate the inference process of 25 randomly selected training samples. And exclude itself from the nearest neighbor search in the dual memory bank during the inference process. The purpose of the above operation is to get the weights and biases about the mapping function. The specific computational procedure is shown below:

$$w = \frac{sdf_{upper} - sdf_{lower}}{rgb_{upper} - rgb_{lower}} \tag{20}$$

$$b = 3d_{lower} - w \times rgb_{lower} \tag{21}$$

where w is the weight of the mapping function, and b is the bias. sdf_{upper} and sdf_{lower} are the 3-sigma ranges of the SDF distribution (mean $\pm$ 3 standard deviations). rgb_{upper} and rgb_{lower} are the 3-sigma ranges of the RGB distribution. The process of applying the mapping function to the RGB score distribution is described as follows:

$$s'\_map_{rgb} = s\_map_{rgb} \times w + b \tag{22}$$

where s'_map_{rgb} represents the new RGB score map after mapping. Finally, the scores of the score maps for each modality are compared pixel by pixel and the larger value at each pixel position is selected as the pixel-level anomaly score. The above process is shown below:

$$s\_map(p)_{finall} = \max\left[s'\_map(p)_{rgb}, s\_map(p)_{sdf}\right] \tag{23}$$

where s_map_{finall} represents the final anomaly localization result after combining the advantages of the two modalities. In addition, we perform Gaussian blurring on all the anomaly score maps involved in the above process to reduce noise and make the score maps smoother. We sort the anomaly scores of all pixels in the score map and use these values as candidate thresholds to compute precision and recall, respectively. The F1 score is then used to evaluate the balance between precision and recall. The threshold corresponding to the highest F1 score is selected as the final segmentation threshold. All pixels with anomaly scores higher than this threshold are identified as anomalous regions.

Moreover, compared with using a single modality, incorporating both RGB and 3D data increases the model complexity and memory usage. Specifically, when using only RGB data, the proposed method requires FLOPs of 5.02 G and memory of 1742 MB. When using only 3D data, it requires FLOPs of 1.29 G and memory of 2104 MB. When both modalities are used, the method requires FLOPs of 6.31 G and memory of 3820 MB. In practical industrial scenarios, especially with large-scale datasets, it is important to make a trade-off between computational cost and detection performance. Flexible modality combinations can further enhance the effectiveness of the proposed method.

IV. Experiments

A. Dataset and Evaluation Metrics

1) Dataset: We evaluate the effectiveness of the proposed method on two multimodal anomaly detection datasets. The MVTec 3D-AD [15] dataset contains 10 different types of industrial objects, totaling 2656 training samples, 294 validation samples, and 1197 test samples. The Eyecandies [46] dataset is a synthetic dataset containing 10 categories of candies, with a total of 10000 training samples, 1000 validation samples, and 4000 test samples. Each sample in these datasets provides 3D information that is precisely aligned with the corresponding RGB pixels, allowing us to obtain the (x, y, z) coordinates at each pixel. This spatial information, combined with the corresponding color data, enables us to effectively perform the detection task. In this paper, we focus on utilizing the complementary relationship between RGB and 3D information to enhance anomaly detection performance.

2) Evaluation Metrics: In this study, multiple evaluation metrics are used to comprehensively assess the anomaly detection and segmentation performance of the model. Anomaly detection performance is evaluated by calculating the area under the receiver operating characteristic curve (I-AUROC), with higher I-AUROC values indicating better overall detection performance. Segmentation performance is evaluated using the pixel-level area under the receiver operating characteristic curve (P-AUROC) and the area under the per-region overlap curve (AUPRO) [47]. AUPRO assesses how well the predicted regions align with the actual anomaly regions. Typically, an integration threshold of 0.3 for the false positive rate (FPR) is used to calculate AUPRO (referred to as AUPRO@30%). To reduce the FPR in industrial applications and meet the demands of high-precision scenarios, this study also employs AUPRO with a more stringent 0.01 integration threshold (referred to as AUPRO@1%).

B. Implementation Details

1) Data Preprocess: The preprocessing of point clouds involves two main steps: background removal and local patch extraction. First, background point clouds in the raw data, which interfere with modeling and analysis, are removed based on the BTF method [14]. Then, since recent advances [44], [48] in analyzing and modeling point clouds by dividing them into multiple local regions have performed well, this study utilizes farthest-point sampling (FPS) [49] to sample a set of points from the original cloud. For each sampled point, we identify its K nearest neighbors within the receptive field to form local patches. In addition, the resolution of the original point cloud and images is resized to 224×224 using nearest-neighbor and bicubic interpolation [14]. To ensure full coverage of the point cloud while allowing patches to share some neighborhood information, local patches are set to be overlappable.

2) Parameter Settings: To ensure that overlapping local patches collectively cover all points in the point cloud sample, each 3D patch contains 500 points (K = 500), and the overlap rate is set to 10. For a point cloud sample with approximately 7,750 points, this results in around 155 local patches, calculated as "Overlap Rate $\times$ Total Number of Points $\div$ K." During SDF model training, following the same settings as PCP [44], we sample 20 query points around each real point. The learning rate and batch size are set to 0.0001 and 32, respectively. All experiments are conducted on a single NVIDIA GeForce RTX 3090, using the PyTorch 1.7.1 deep learning framework.

3) Training Process: The 3D signed distance function (SDF) model employs pulling loss to minimize the distance between pulled points and actual surface points. This enables the neural implicit function (NIF) model to learn to predict signed distance values, thereby achieving an implicit representation of local geometric information. The SDF features extracted from normal point clouds by PointNet are stored in the memory bank M_{3D}. The RGB features of normal samples extracted by Wide-ResNet50-2 are refined through the attention mechanism and then stored in memory bank M_{2D} under the guidance of SDF features. Together, M_{3D} and M_{2D} form the attention-enhanced dual memory bank.

4) Inference Process: During inference, for each SDF feature, we search for its 10 nearest neighbors in M_{3D}. These neighbors serve as basis vectors to reconstruct the input SDF feature using sparse coding. The reconstructed feature is then fed into the NIF model, where the pretrained SDF model computes the signed distance values for each local patch. The SDF anomaly score map is obtained using the designed weighted map anomaly scoring scheme. In M_{2D}, all RGB features corresponding to the SDF features used for reconstruction in M_{3D} form a memory-guided RGB dictionary. For each input RGB feature, we find its nearest neighbor in this dictionary and reconstruct it using sparse coding. The RGB anomaly score map is then obtained using the cosine similarity anomaly scoring scheme. Finally, the RGB scores are mapped to the 3D score distribution. At each pixel location, the higher score between the two modalities is selected as the final anomaly response.

C. Experiments Analysis

In this section, we provide a detailed comparison of the proposed method with other state-of-the-art approaches under different modal information as input. AST [19] effectively reduces

TABLE II
IMAGE-LEVEL AUROC (I-AUROC) RESULTS OF VARIOUS METHODS FOR ANOMALY DETECTION PERFORMANCE ON THE MVTEC 3D-AD DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

	Method	Bagel	Cable Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
3D	Depth GAN [15]	0.530	0.376	0.607	0.603	0.497	0.484	0.595	0.489	0.536	0.521	0.523
	Depth AE [15]	0.468	0.731	0.497	0.673	0.534	0.417	0.485	0.549	0.564	0.546	0.549
	DepthVM [15]	0.510	0.542	0.469	0.576	0.609	0.699	0.450	0.419	0.668	0.520	0.546
	Voxel GAN [15]	0.383	0.623	0.474	0.639	0.564	0.409	0.617	0.427	0.663	0.577	0.537
	Voxel AE [15]	0.693	0.425	0.515	0.790	0.494	0.558	0.537	0.484	0.639	0.583	0.571
	Voxel VM [15]	0.750	**0.747**	0.613	0.738	0.823	0.693	0.679	0.652	0.609	0.690	0.699
	3D-ST [18]	0.862	0.484	0.832	0.894	0.848	0.663	0.763	0.687	0.958	0.486	0.748
	BTF (FPFH) [14]	0.820	0.533	0.877	0.769	0.718	0.574	0.774	0.895	**0.990**	0.582	0.753
	AST [19]	0.881	0.576	0.965	0.957	0.679	**0.797**	0.990	0.915	0.956	0.611	0.833
	Shape-guided [22]	0.983	0.682	**0.978**	0.998	**0.960**	0.737	**0.993**	**0.979**	0.966	0.871	0.916
	M3DM [21]	0.941	0.651	0.965	0.969	0.905	0.760	0.880	0.974	0.926	0.765	0.874
	Ours	**0.985**	0.678	0.972	**0.999**	0.950	0.720	**0.993**	0.969	0.972	**0.949**	**0.919**
RGB	PADiM [11]	**0.975**	0.775	0.698	0.582	0.959	0.663	0.858	0.535	0.832	0.760	0.764
	PatchCore [12]	0.876	0.880	0.791	0.682	0.912	0.701	0.695	0.618	0.841	0.702	0.770
	CFLOW [28]	0.880	0.858	0.828	0.563	**0.986**	0.738	0.757	0.628	0.970	0.720	0.793
	BTF (RGB) [14]	0.854	0.840	0.824	0.687	0.974	0.716	0.713	0.593	0.920	0.724	0.785
	AST [19]	0.947	0.928	0.851	**0.825**	0.981	**0.951**	0.895	0.613	**0.992**	**0.821**	**0.880**
	Shape-guided [22]	0.911	0.936	0.883	0.662	0.974	0.772	0.785	0.641	0.884	0.706	0.815
	M3DM [21]	0.944	0.918	**0.896**	0.749	0.959	0.767	**0.919**	0.648	0.938	0.767	0.850
	Ours	0.916	**0.974**	0.836	0.625	0.973	0.828	0.778	**0.682**	0.913	0.742	0.827
RGB+3D	Depth GAN [15]	0.538	0.372	0.580	0.603	0.430	0.534	0.642	0.601	0.443	0.577	0.532
	Depth AE [15]	0.648	0.502	0.650	0.488	0.805	0.522	0.712	0.529	0.540	0.552	0.595
	DepthVM [15]	0.513	0.551	0.477	0.581	0.617	0.716	0.450	0.421	0.598	0.623	0.555
	Voxel GAN [15]	0.680	0.324	0.565	0.399	0.497	0.482	0.566	0.579	0.601	0.482	0.517
	Voxel AE [15]	0.510	0.540	0.384	0.693	0.446	0.632	0.550	0.494	0.721	0.413	0.538
	Voxel VM [15]	0.553	0.772	0.484	0.701	0.751	0.578	0.480	0.466	0.689	0.611	0.609
	3D-ST [18]	0.950	0.483	**0.986**	0.921	0.905	0.632	0.945	**0.988**	0.976	0.542	0.833
	BTF [14]	0.938	0.765	0.972	0.888	0.960	0.664	0.904	0.929	0.982	0.726	0.873
	AST [19]	0.983	0.873	0.976	0.971	0.932	0.885	0.974	0.981	**1.000**	0.797	0.937
	Shape-guided [22]	0.986	0.894	0.983	0.991	0.976	0.857	**0.990**	0.965	0.960	0.869	0.947
	M3DM [21]	**0.994**	0.909	0.972	0.976	0.960	**0.942**	0.973	0.899	0.972	0.850	0.945
	MIAD [20]	**0.994**	0.888	0.984	**0.993**	0.980	0.888	0.941	0.943	0.980	**0.953**	0.954
	MIAD-M [20]	0.988	0.875	0.984	0.992	**0.997**	0.924	0.964	0.949	0.979	0.950	**0.960**
	MIAD-S [20]	0.983	0.878	0.973	0.992	0.987	0.913	0.900	0.936	0.981	0.941	0.948
	MIAD-T [20]	0.948	0.784	0.946	0.985	0.946	0.855	0.815	0.932	0.989	0.794	0.899
	Ours	0.988	**0.975**	0.978	0.977	0.976	0.889	0.976	0.952	0.966	0.878	0.956

the overgeneralization problem for anomalous samples but fails to fully leverage the advantages of 3D data in geometric anomaly detection. This limits its performance in multimodal scenarios. MIAD [20] achieves anomaly detection by learning two cross-modal mapping functions, but it requires designing complex training objectives. MIAD-M, MIAD-S, and MIAD-T are three variants obtained by pruning the network layers of the feature extractor in MIAD. BTF [14] improves the utilization of multimodal data by connecting features from two modalities, but it overlooks the potential interference between the modalities. M3DM [21] strengthens single-domain reasoning ability by constructing three memory banks but leads to significant memory consumption. Shape-guided [22] reduces memory usage by using dual memory banks but neglects the quality of features in the memory banks.

Table II presents the anomaly detection performance using I-AUROC, while Table III highlights the anomaly segmentation performance measured by AUPRO. Additionally, Table VI provides a comparative analysis of segmentation performance, focusing specifically on P-AUROC. **1)** The proposed method achieves the highest I-AUROC, outperforming M3DM [21] by 4.5% when using only 3D information for anomaly detection. This result highlights the effectiveness of learning from overlapping local patches through PointNet [43]. For the segmentation task, the method achieves the best AUPRO in six data subclasses and the second highest in four others, demonstrating its strong generalization across different data categories, which is essential for practical industrial applications. **2)** In anomaly detection using RGB information, since the proposed method focuses on effectively combining multimodal information, i.e., RGB information guided by 3D information is used. However, compared to AST [19], which directly uses RGB features for detection, the proposed method shows slightly lower performance in I-AUROC. In contrast, for the segmentation task, the proposed approach achieves optimal results with 94.4% AUPRO, attributed to multi-dimensional attention focused on learning the details of RGB features and the application of cosine similarity for sparse RGB feature computation. **3)** In multimodal anomaly detection using RGB + 3D, the proposed method achieves a second-best I-AUROC of 95.6% , surpassing AST by 1.9% , and outperforming BTF [14] and M3DM, both of which utilize memory banks, by 8.3% and 1.1% , respectively. In terms of segmentation performance, the proposed method also attains optimal results. These results highlight the advantages of the

TABLE III
AUPRO@30% Results of Various Methods for Anomaly Segmentation Performance on the MVTec 3D-AD Dataset

	Method	Bagel	Cable Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
3D	Depth GAN [15]	0.111	0.072	0.212	0.174	0.160	0.128	0.003	0.042	0.446	0.075	0.143
	Depth AE [15]	0.147	0.069	0.293	0.217	0.207	0.181	0.164	0.066	0.545	0.142	0.203
	DepthVM [15]	0.280	0.374	0.243	0.526	0.485	0.314	0.199	0.388	0.543	0.385	0.374
	Voxel GAN [15]	0.440	0.453	0.875	0.755	0.782	0.378	0.392	0.639	0.775	0.389	0.583
	Voxel AE [15]	0.260	0.341	0.581	0.351	0.502	0.234	0.351	0.658	0.015	0.185	0.348
	Voxel VM [15]	0.453	0.343	0.521	0.697	0.680	0.284	0.349	0.634	0.616	0.346	0.492
	EasyNet [40]	0.160	0.030	0.680	0.759	0.758	0.069	0.225	0.734	0.797	0.509	0.472
	M3DM [21]	0.943	**0.818**	**0.977**	0.882	**0.881**	**0.743**	0.958	0.974	0.950	0.929	0.906
	Ours	**0.969**	0.770	0.974	**0.932**	0.872	0.705	**0.976**	**0.983**	**0.951**	**0.942**	**0.907**
RGB	CFLOW [28]	0.855	0.919	0.958	0.867	**0.969**	0.500	0.889	0.935	0.904	0.919	0.871
	PatchCore [12]	0.901	0.949	0.928	0.877	0.892	0.563	0.904	0.932	0.908	0.906	0.876
	EasyNet [40]	0.751	0.825	0.916	0.599	0.698	0.699	0.917	0.827	0.887	0.636	0.776
	BTF (RGB) [14]	0.898	0.948	0.927	0.872	0.927	0.555	0.902	0.931	0.903	0.899	0.876
	Shape-guided [22]	0.946	0.972	0.960	**0.914**	0.958	0.776	0.937	0.949	0.956	0.957	0.933
	M3DM [21]	0.952	0.972	**0.973**	0.891	0.932	**0.843**	**0.970**	0.956	**0.968**	**0.966**	0.942
	Ours	**0.965**	**0.977**	0.967	0.912	0.966	0.827	0.950	**0.960**	0.959	0.960	**0.944**
RGB+3D	Depth GAN [15]	0.421	0.422	0.778	0.696	0.494	0.252	0.285	0.362	0.402	0.631	0.474
	Depth AE [15]	0.432	0.158	0.808	0.491	0.841	0.406	0.262	0.216	0.716	0.478	0.481
	DepthVM [15]	0.388	0.321	0.194	0.570	0.408	0.282	0.244	0.349	0.268	0.331	0.335
	Voxel GAN [15]	0.664	0.620	0.766	0.740	0.783	0.332	0.582	0.790	0.633	0.483	0.639
	Voxel AE [15]	0.467	0.750	0.808	0.550	0.765	0.473	0.721	0.918	0.019	0.170	0.564
	Voxel VM [15]	0.510	0.331	0.413	0.715	0.680	0.279	0.300	0.507	0.611	0.366	0.471
	3D-ST [18]	0.950	0.483	**0.986**	0.921	0.905	0.632	0.945	**0.988**	0.976	0.542	0.833
	BTF [14]	0.976	0.967	0.979	**0.974**	**0.971**	0.884	0.976	0.981	0.959	0.971	0.964
	AST [19]	0.970	0.947	0.981	0.939	0.913	0.906	0.979	0.982	0.889	0.940	0.944
	Shape-guided [22]	**0.981**	0.973	0.982	0.971	0.962	**0.978**	0.981	0.983	0.974	0.975	**0.976**
	M3DM [21]	0.970	0.971	0.979	0.950	0.941	0.932	0.977	0.971	0.971	0.975	0.964
	MIAD [20]	0.979	0.972	0.982	0.945	0.950	0.968	0.980	0.982	0.975	0.981	0.971
	MIAD-M [20]	0.980	0.966	0.982	0.947	0.959	0.967	**0.982**	0.983	0.976	**0.982**	0.972
	MIAD-S [20]	0.978	0.960	0.982	0.948	0.960	0.972	0.977	0.983	0.976	0.981	0.972
	MIAD-T [20]	0.977	0.903	0.981	0.950	0.945	0.956	0.973	0.983	0.973	0.973	0.961
	Ours	**0.981**	**0.978**	0.981	0.972	0.962	0.977	0.981	0.983	**0.978**	0.968	**0.976**

TABLE IV
AUPRO@1% Results of Various Methods for Anomaly Segmentation Performance on the MVTec 3D-AD Dataset

Method	Bagel	Cable Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
BTF [14]	0.428	0.365	0.452	0.431	0.370	0.244	0.427	0.470	0.298	0.345	0.383
AST [19]	0.388	0.322	0.470	0.411	0.328	0.275	0.474	0.487	0.360	0.474	0.398
Shape-guided [22]	**0.481**	0.396	0.482	0.468	0.387	**0.449**	0.483	**0.504**	0.448	0.446	0.454
M3DM [21]	0.414	0.395	0.447	0.318	**0.422**	0.335	0.444	0.351	0.416	0.398	0.394
MIAD [20]	0.459	0.431	0.485	0.469	0.394	0.413	0.468	0.487	0.464	0.476	0.455
MIAD-M [20]	0.480	0.398	0.490	0.467	0.413	0.408	0.481	0.494	0.468	**0.488**	0.459
MIAD-S [20]	0.461	0.379	**0.492**	0.479	0.411	0.429	0.430	0.494	0.467	0.472	0.451
MIAD-T [20]	0.449	0.267	0.487	**0.487**	0.364	0.369	0.395	0.491	0.462	0.421	0.419
Ours	0.478	**0.449**	0.477	0.471	0.405	0.444	**0.489**	0.498	**0.473**	0.429	**0.461**

attention-enhanced memory-guided network, while demonstrating the effectiveness of the strategy that combines the strengths of each modality's detection based on memory-guided conditions. Table V presents the results of various multimodal methods on the Eyecandies dataset across multiple metrics. The results show that the proposed method achieves the best anomaly segmentation performance, demonstrating its strong generalization ability across different detection targets.

Table IV demonstrates the actual performance of each method in high precision requirement scenarios. The requirement of low FPR has been the focus of industrial applications. The proposed method achieves optimal and sub-optimal AUPRO@1% on 5 classes of data from MVTec 3D-AD, and the overall performance is 7.8% higher than that of BTF, 6.7% higher than that of M3DM, and 0.6% higher than that of the current state-of-the-art MIAD [20]. This study provides a booster study for the development of anomaly detection in high-precision industrial scenarios.

Fig. 5 presents a comparison of anomaly localization results between the proposed method and Shape-guided [22]. The proposed method shows superior performance in reducing both over-detection and missed detection, as illustrated by the score and segmentation maps. Specifically, the Shape-guided method exhibits multiple instances of over-detection in the second row and a few minor cases of over-detection in the fourth row, which the proposed method effectively mitigates. The results in the third row highlight the advantage of the proposed method in minimizing missed detections. Overall, the Shape-guided method has two weaknesses. **1)** The normal regions around the target often have high anomaly scores. The proposed method

TABLE V
MULTI-METRIC RESULTS OF VARIOUS MULTIMODAL METHODS ON THE EYECANDIES DATASET

	Method	Can. C.	Cho. C.	Cho. P.	Conf.	Gum. B.	Haz. T.	Lic. S.	Lollip.	Marsh.	Pep. C.	Mean
I-AUROC	AST [19]	0.574	0.747	0.747	0.889	0.596	0.617	0.816	0.841	0.987	0.987	0.780
	Shape-guided [22]	0.634	0.973	0.803	0.920	0.758	0.645	0.920	0.860	0.990	0.976	0.848
	M3DM [21]	0.624	0.958	**0.958**	**1.000**	**0.886**	0.758	**0.949**	0.836	**1.000**	**1.000**	**0.897**
	MIAD [20]	**0.680**	0.931	0.952	0.880	0.865	**0.782**	0.917	0.840	0.998	0.962	0.881
	MIAD-M [20]	0.645	0.936	0.914	0.901	0.845	0.747	0.877	**0.904**	0.992	0.885	0.865
	Ours	0.602	**0.995**	0.810	0.981	0.798	0.736	0.918	0.881	0.995	0.984	0.870
AUPRO@30%	AST [19]	0.514	0.835	0.714	0.905	0.587	0.590	0.736	0.769	0.918	0.878	0.744
	Shape-guided [22]	0.881	0.916	0.728	0.953	0.843	0.624	0.917	**0.949**	0.969	0.938	0.872
	M3DM [21]	0.906	0.923	0.803	**0.983**	0.855	0.688	0.880	0.906	0.966	0.955	0.882
	MIAD [20]	0.942	0.902	**0.831**	0.965	**0.875**	0.762	0.791	0.913	0.939	0.946	0.887
	MIAD-M [20]	**0.943**	0.892	0.795	0.962	0.871	**0.779**	0.767	0.909	0.944	0.935	0.880
	Ours	0.910	**0.924**	0.734	0.967	0.873	0.689	**0.927**	0.938	**0.970**	**0.959**	**0.889**
AUPRO@1%	AST [19]	0.035	0.230	0.129	0.234	0.092	0.069	0.139	0.090	0.255	0.224	0.149
	Shape-guided [22]	0.076	0.352	0.245	0.387	0.245	0.079	0.320	**0.318**	0.436	0.395	0.285
	M3DM [21]	0.166	0.388	0.329	0.486	0.315	0.131	0.323	0.258	**0.462**	**0.454**	0.331
	MIAD [20]	**0.229**	0.397	**0.345**	0.389	**0.353**	0.188	0.333	0.236	0.455	0.428	**0.335**
	MIAD-M [20]	0.223	0.389	0.333	0.395	0.348	**0.206**	**0.342**	0.225	0.452	0.385	0.330
	Ours	0.143	**0.416**	0.290	**0.492**	0.343	0.136	0.334	0.314	0.441	0.437	**0.335**

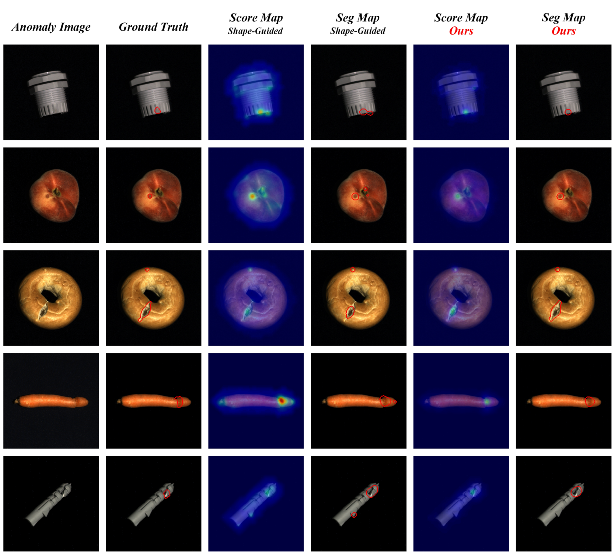


Fig. 5. Qualitative results of the proposed method and other methods on the MVTec 3D-AD dataset. From left to right: RGB anomaly image, ground truth, score map and segmentation map of the Shape-guided method, and score map and segmentation map of our method.

addresses this issue using an anomaly scoring scheme based on a weight map of signed distance values. This effectively highlights anomalous regions while keeping anomaly scores lower in normal areas. **2**) There are over-detection and missed detection issues. Our method enhances the informativeness and discriminability of feature descriptors through an attention mechanism. This significantly improves the quality of data stored in the memory bank and helps mitigate these issues. More anomaly localization results for the proposed method are provided in Fig. 6.

D. Frame Rate and Memory Occupancy

In Table VI, we provide a detailed comparison of the proposed method with previous advanced methods in terms of memory footprint and inference speed (frames per second). In the field of AD, methods that utilize memory banks have attracted much attention due to the fact that accurate localization of anomalies can be achieved with only normal samples. However, these methods depend heavily on the quality of the data used to construct the memory bank. Performance can be adversely affected if the input data is of low quality or contains noise and bias. This study performs a series of preprocessing steps on point cloud data to reduce potential interference. By learning and storing local features of sample data, it achieves a more detailed representation of normal features, thereby enhancing the data quality of the memory bank. In addition, previous methods tend to identify anomalies by calculating the similarity between each anomalous sample and all normal samples when using a memory bank, which can bring a huge computational burden. The attention-enhanced memory-guided network proposed in this study addresses this by only using the K most relevant nearest-neighbor features from the memory bank for each anomaly feature, thereby improving inference efficiency.

Compared to the M3DM method, which uses three memory banks, the proposed method reduces memory usage by 2706.12 MB and increases inference speed by 0.118 fps. The BTF method attempts to integrate 3D features into the 2D features provided by a frozen convolutional model (Wide-ResNet50-2) [50], but this approach fails to fully leverage the benefits of multimodality. The AST method, based on normalized flow, employs a teacher-student architecture that favors faster inference; however, it uses 3D data only as an additional input channel in a 2D network architecture, resulting in lower performance than M3DM. MIAD detects anomalies by evaluating inconsistencies between learned features from feature extractors and cross-modal mapped features, achieving faster inference speeds. Compared with all these methods, the proposed approach outperforms others across various metrics, including I-AUROC, P-AUROC, AUPRO@30% , and the more stringent AUPRO@1% . Fig. 7(a) and (b) clearly highlight the advantages demonstrated by the proposed approach in anomaly detection and anomaly segmentation. Importantly, the proposed algorithm surpasses the previous state-of-the-art method MIAD in the AD task. Moreover, our work can be reproduced using an

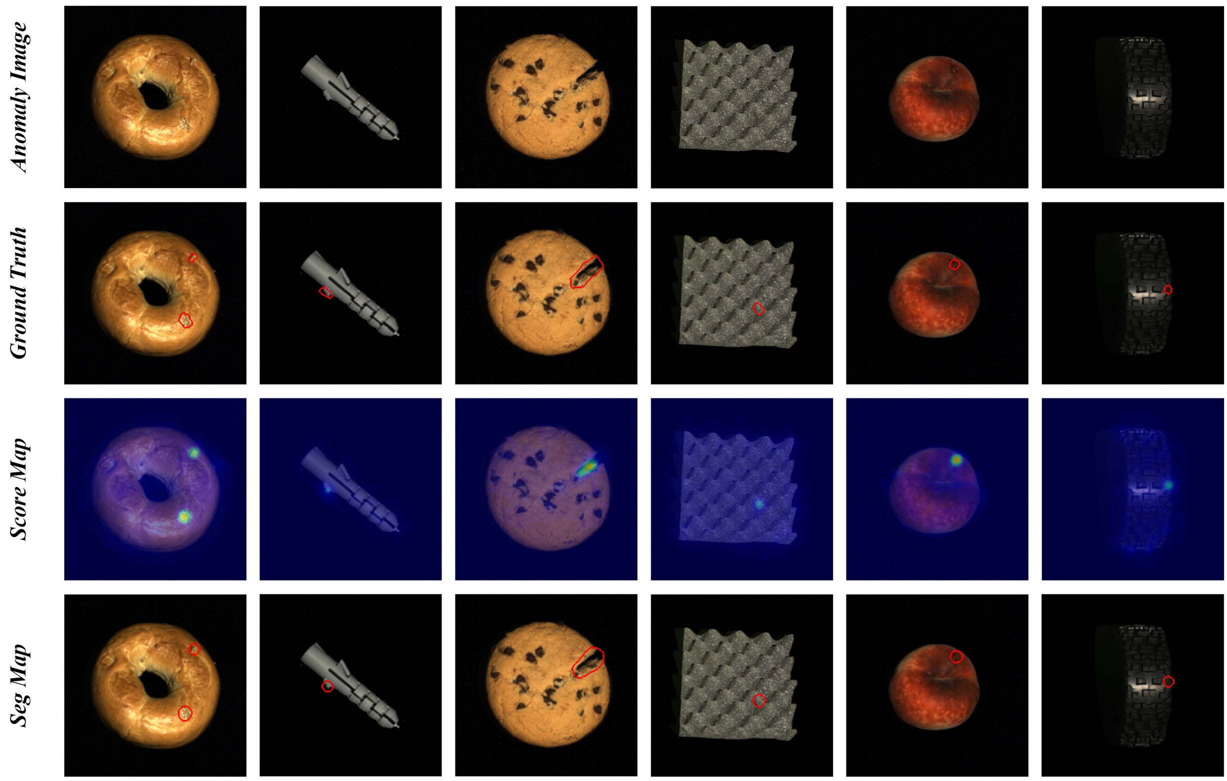
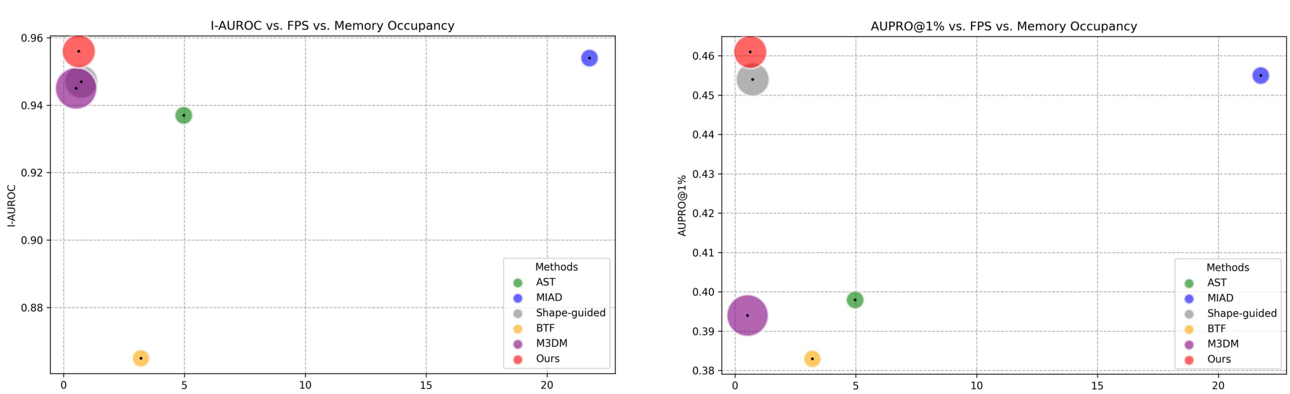


Fig. 6. Anomaly localization results of the proposed method on the MVTec 3D-AD dataset. From top to bottom: RGB anomaly image, ground truth, and the score map and segmentation map of our method.

TABLE VI
INFERENCE SPEED, MEMORY OCCUPANCY, AND AD PERFORMANCE ON THE MVTEC 3D-AD DATASET. INFERENCE SPEED IS MEASURED IN FPS, AND MEMORY IN MB.

Method	FPS	Memory Occupancy	I-AUROC	P-AUROC	AUPRO@30%	AUPRO@1%
BTF [14]	3.197	**381.06**	0.865	0.992	0.964	0.383
AST [19]	4.966	463.94	0.937	0.976	0.944	0.398
Shape-guided [22]	0.728	3826.00	0.947	**0.996**	**0.976**	0.454
M3DM [21]	0.514	6526.12	0.945	0.992	0.964	0.394
MIAD [20]	**21.755**	437.91	0.954	0.993	0.971	0.455
Ours	0.632	3820.00	**0.956**	**0.996**	**0.976**	**0.461**



(a) (b)

Fig. 7. Comprehensive comparison of multimodal anomaly detection methods in terms of anomaly detection performance (I-AUROC, as shown in subfigure (a)), anomaly segmentation performance (AUPRO@1% , as shown in subfigure (b)), inference speed, and memory occupancy. The area of each circle represents memory usage.

TABLE VII
ABLATION RESULTS OF EACH COMPONENT OF THE PROPOSED METHOD ON THE MVTEC 3D-AD DATASET IN TERMS OF I-AUROC, AUPRO@30% , AND AUPRO@1% .

L2	Cosine	Weight Map	Attention	I-AUROC	AUPRO@30%	AUPRO@1%
✓		✓		0.947	0.972	0.444
	✓	✓		0.950	0.974	0.452
	✓		✓	**0.956**	0.974	0.450
✓		✓	✓	0.955	0.970	0.438
	✓	✓	✓	**0.956**	**0.976**	**0.461**

TABLE VIII
ABLATION RESULTS ON THE EFFECTIVENESS OF THE THREE ATTENTION BRANCHES ON THE MVTEC 3D-AD DATASET

Attention	I-AUROC	AUPRO@30%	AUPRO@1%
Channel	0.954	0.973	0.456
Height	0.953	0.975	0.453
Width	0.950	0.972	0.452
Height + Width	0.954	0.975	0.458
Channel + Height + Width	**0.956**	**0.976**	**0.461**

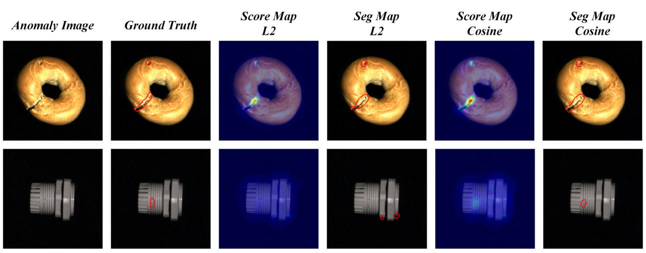


Fig. 8. Qualitative results of different RGB anomaly score calculation methods.

NVIDIA GeForce RTX 3090, and more powerful hardware is expected to further improve the frame rate.

E. Ablation Study

This section presents a detailed ablation study. Table VII highlights the advantage of using cosine similarity over L2 distance for anomaly scoring. It also highlights the benefits of the weighted map scoring scheme for anomaly segmentation and the role of the attention mechanism in improving the quality of feature data stored in the memory bank. Table VIII provides a more detailed ablation study on the three attention branches. Additionally, we present a more intuitive visual comparison of ablation studies. Fig. 8 demonstrates the scoring superiority of cosine similarity over L2 distance. By enhancing the representation of RGB features, the attention mechanism also improves the data quality within the memory bank. Segmentation results in Fig. 9 illustrate the attention mechanism's significant contribution to detecting subtle anomalies (W/O represents without, W/ represents with). Fig. 10 compares the effectiveness of directly using the absolute value of SDF as the anomaly score versus the weight map approach for anomaly localization. Table IX provides a detailed analysis of patch sizes. While K=300 yields the highest I-AUROC, K=500 offers the optimal balance between inference speed and anomaly segmentation accuracy. In fact, as the patch size increases, the inference efficiency of the proposed method gradually improves. However, this comes at the cost of reduced model performance. For industrial deployment, the patch size can be adjusted flexibly to adapt to the specific requirements of different tasks.

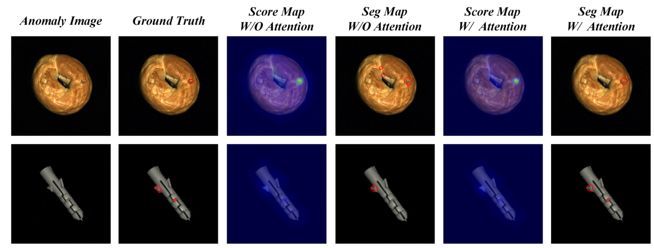


Fig. 9. Ablation study of the attention mechanism.

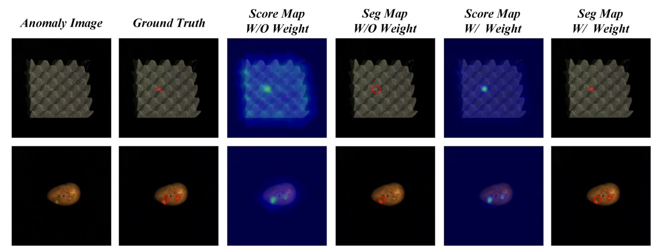


Fig. 10. Ablation study of the weight map.

TABLE IX
INFERENCE TIME, ANOMALY DETECTION PERFORMANCE, AND ANOMALY SEGMENTATION PERFORMANCE WITH VARIOUS PATCH SIZE K SETTINGS ON THE MVTEC 3D-AD DATASET

K	I-AUROC	AUPRO@30%	AUPRO@1%	Inference Time
700	0.939	0.971	0.453	**1.36**
600	0.946	0.973	0.456	1.42
500	0.956	**0.976**	**0.461**	1.58
400	0.956	0.975	0.456	1.74
300	**0.961**	0.975	0.458	2.06

TABLE X
ANOMALY DETECTION AND SEGMENTATION PERFORMANCE OF THE PROPOSED METHOD UNDER FEW-SHOT SETTINGS ON THE MVTEC 3D-AD DATASET

Method	I-AUROC	P-AUROC	AUPRO@30%	AUPRO@1%
5-shot	0.786	0.986	0.942	0.344
10-shot	0.813	0.989	0.953	0.381
50-shot	0.891	0.994	0.969	0.425
Full	0.956	0.996	0.976	0.461

F. Performance of Few-Shot Setting

We evaluate the proposed method under few-shot settings, and the results are shown in Tables X. Specifically, we randomly select 5, 10, and 50 images from each category of the MVTec 3D-AD dataset as training data. The model is then tested on the full test set to assess its anomaly detection and segmentation performance under few-shot conditions. Notably, in the 5-shot and 10-shot settings, our method still outperforms some approaches that are not designed for few-shot scenarios. The few-shot nature enables our method to better scale to practical applications.

TABLE XI
ABLATION RESULTS OF THE PROPOSED METHOD'S MULTIMODAL DATA INPUT ARCHITECTURE ON AUPRO@30%

Method	Bagel	Cable Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
3D	0.969	0.770	0.974	0.932	0.872	0.705	0.976	**0.983**	0.951	0.942	0.907
RGB	0.965	0.977	0.967	0.912	**0.966**	0.827	0.950	0.960	0.959	0.960	0.944
3D + RGB	**0.981**	**0.978**	**0.981**	**0.972**	0.962	**0.977**	**0.981**	**0.983**	**0.978**	**0.968**	**0.976**

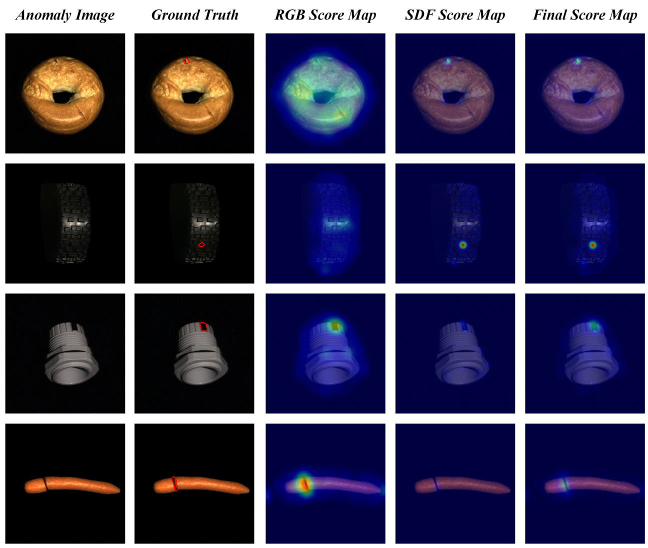


Fig. 11. Qualitative results before and after integrating multimodal information. From left to right: anomaly image, ground truth, score map and segmentation map from the RGB branch, score map and segmentation map using only 3D information, and score map and segmentation map after combining RGB and 3D. The results show that our method solves the problem of anomaly localization failure caused by relying solely on a single modality.

TABLE XII
COMPARISON OF THE COMPUTATIONAL COMPLEXITY, INFERENCE SPEED, AND MEMORY OCCUPANCY OF THE PROPOSED METHOD USING SINGLE-MODAL AND MULTIMODAL DATA AS INPUT ON THE MVTEC 3D-AD DATASET

Method	FLOPs	FPS	Memory Occupancy
RGB	5.02	3.667	1742.00
3D	1.29	0.719	2104.00
RGB+3D	6.31	0.632	3820.00

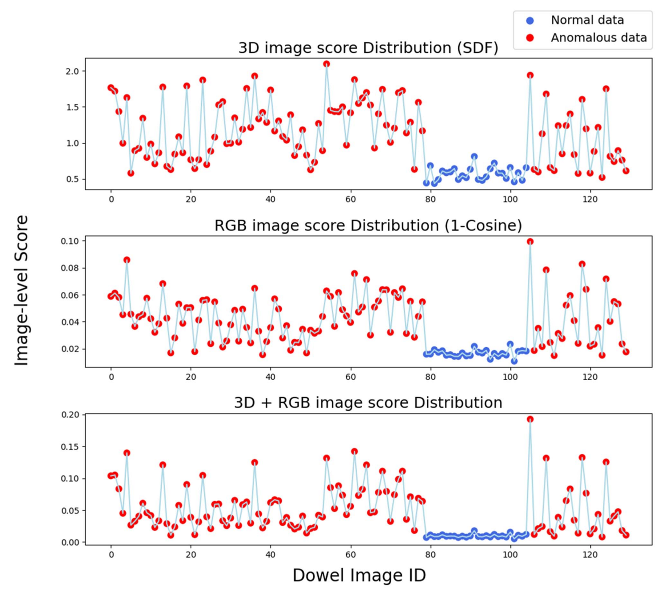


Fig. 12. Distribution of anomaly detection scores before and after fusion on the dowel subclass. The relative difference in image-level scores between normal and abnormal data is enhanced after the 3D + RGB fusion, facilitating better anomaly detection.

G. Effectiveness of Multimodal Integration

This section provides a detailed analysis of the effectiveness of combining multimodal data. Fig. 11 illustrates the benefits of combining RGB and 3D information in the proposed method, which effectively avoids the problem of failing to localize anomalies in the case of a single modality. The types of anomalies are diverse. RGB information tends to work well at identifying color defects in objects, but it does not work well for holes in bagels with complex surfaces. In contrast, 3D information is typically effective for detecting spatial anomalies on the test object. Therefore, effectively combining RGB and 3D information is key to improving anomaly detection and localization performance. This study realizes the effective utilization of multimodal information by merging the detection advantages in the two modalities. Table XI presents AUPRO@30% results for each single modality and after combining the multimodal information. Table XII shows the computational complexity, inference speed, and memory occupancy of the models using single modal and multimodal data respectively. The results indicate that, compared to single modal data, although multimodal data requires more memory occupancy, it achieves superior performance by integrating the advantages of each modality. Fig. 12 shows the distribution of image-level scores for the "Dowel" subclass under various modality conditions. In normal data, scores should ideally be lower than those of anomalous data. After combining RGB and 3D information, the relative size of the score distributions for normal and abnormal data is enlarged. This will help to better identify anomalies.

V. CONCLUSION AND LIMITATIONS

In this paper, we propose a multimodal industrial anomaly detection method based on an attention-enhanced memory-guided network. It leverages point cloud and RGB image data to provide a more comprehensive and accurate solution for quality control in automated production lines. By designing an attention-enhanced dual memory bank, the single-domain inference ability of each modality is fully preserved. The score distributions from different modalities are fused through score mapping, effectively combining the strengths of point cloud data for geometric anomaly detection and RGB data for color anomaly detection.

Additionally, we propose an anomaly scoring scheme based on a weight map of signed distance values, which further improves segmentation precision of the model. On the MVTec 3D-AD and Eyecandies datasets, our method achieves state-of-the-art anomaly segmentation performance compared to other advanced approaches. Even under stricter false positive rate settings, our method maintains superior segmentation performance. In the future, with the advancement of computing power and the increasing availability of multimodal data acquisition technologies, multimodal anomaly detection is expected to be widely applied in various industrial fields, driving the development of industrial quality inspection technologies.

Although the proposed method exhibits excellent AD performance, it still has some limitations. Memory-based approaches to implement anomaly detection often bring unavoidable memory footprints, leading to some limitations in inference speed as well. To address these issues, future work will focus on three potential improvement directions. **1)** Dynamic memory pruning: This technique removes memory features with low relevance to the current input during inference. It effectively reduces memory usage while minimizing unnecessary computations. **2)** Lightweight attention module: This module has fewer parameters and lower computational costs. **3)** Knowledge distillation: A lightweight student network, distilled from a teacher network, can be used as the feature extractor.

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